

4.11 Diagnostic Virology I

▼ What does HTLV-1 cause?

- HTLV-1 can cause adult T cell leukaemia (cancer)
- It was the **first virus to be found to cause malignancy**
- Approximately **5% of infected patients will develop cancer**

▼ Why is diagnosis of HTLV-1 infections sometimes difficult?

- HTLV-1 can cause many different diseases with different symptoms, so it can be difficult to identify that HTLV-1 is responsible for symptoms seen in a patient
- These diseases include:
 - Adult T-cell leukaemia
 - Adult T-cell leukaemia/lymphoma
 - HTLV-1-associated myelopathy
 - Tropical spastic paraparesis (TSP)
 - HTLV-1-associated infectious dermatitis
 - HTLV-1-associated uveitis

▼ How can HTLV-1 be transmitted?

- Mother to infant during pregnancy
- Through sexual intercourse
- Blood-to-blood contact (i.e transfusions/sharing needles)

▼ Is HTLV-1 a single or double-stranded virus?

- HTLV-1 is a **single-stranded RNA virus**

▼ What is Tax?

- Tax is a gene that is essential for coding for proteins useful in transcription and viral replication
- Tax is also known to have **severe oncogenic potential by accelerating the transition from the G1 to S phase as well as causing mutations which can lead to cancer.**

▶ How can Tax be used diagnostically?

▶ Which type of cells does HTLV-1 show preferential infection for?

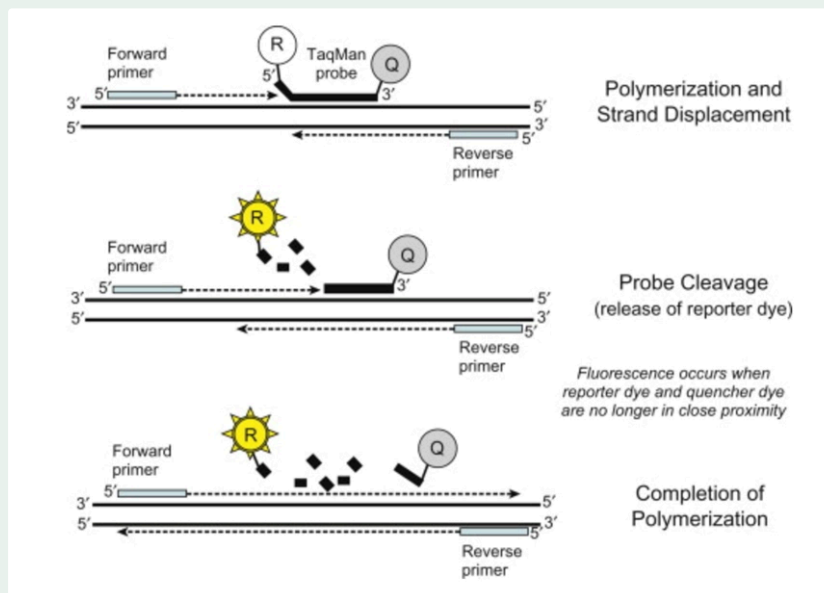
▶ Describe the replication cycle of HTLV-1 from infection.

▶ Why is HTLV-1 easily spread through blood transfusions?

- ▶ What is the purpose of quantitative real-time PCR?
- ▶ What is meant by the statement "clonal proliferation is the dominant contributor to the increase in number of HTLV-1 infected cells rather than de novo infection of cells"?
- ▶ What are peripheral blood mononuclear cells (PBMCs)?
- ▶ How is Ficoll gradient centrifugation of blood samples important in the PCR process?
- ▶ In which direction is DNA synthesised during PCR?
- ▶ Describe the three steps involved in the PCR.
- ▶ How many times are these steps repeated?
- ▶ What is the purpose of the forward and reverse primers in PCR?
- ▶ What is the purpose of the buffer solution in the PCR reaction?
- ▶ What is Taq polymerase and why is it used?
- ▶ What technique is used to visualise the DNA produced from PCR?
- ▶ How are we able to know that the DNA produced from the PCR of PBMC genes are 300 base pairs long?
- ▶ Which two stains can be used to visualise DNA on gel electrophoresis?
- ▶ When can the results of PCR automatically be assumed invalid?
- ▶ What are the five essential components required for PCR mixtures?
- ▶ What are the names of the primers used in PCR for HTLV-1?
- ▶ What is nested PCR?
- ▶ Describe how Western-blotting can be used to determine if a patient has antibodies against HTLV-1.
- ▶ Against which viral proteins are antibodies produced by hosts on the HTLV-1 virus?
- ▶ What is the purpose of quantitative real-time PCR (qrt-PCR)?
- ▶ How does the fluorescent dye in quantitative real-time PCR (qrt-PCR) work?
- ▶ What is the cycle threshold (CT) in qrt-PCR?

▼ What is the Taqman method in qrt-PCR?

- It is an alternative method used instead of the fluorescent dye PCR method which requires the use of an **oligo** to **bind onto the amplifying region** before amplification begins
- The **oligo** consists of both a **fluorescent region** and a **quencher region** and will be **cleaved apart** as the **DNA polymerase moves along the chain**
- This means the **level of fluorescence seen per cycle** will be **proportional to the levels of template DNA** present at each point
 - i.e the **more of the Taq template DNA present in the sample (i.e higher HTLV-1 load)**, the **more oligos will be cleaved by DNA polymerase** and hence the **more fluorescence will be reported**
- This increases the specificity of the reaction, and although it is more expensive, this method (**Taqman method**) is what is commonly used for diagnostic purposes



► Describe how qrt-PCR is used to determine the viral load of a patient.

► **NOTE:** qrt-PCR is sometimes referred to as quantitative real time or quantitative reverse transcriptase PCR, so be careful to clarify nomenclature